

# **FSC Safety Manager Configuration Forms**

**for use with the Honeywell FSC System**

**Release 670**

**FS88-600**



**Implementation  
FSC Safety Manager**

***FSC Safety Manager  
Configuration Forms***

**for use with the Honeywell FSC System**

**Release 670**

**FS88-600**

**06/06**

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## About This Publication

This publication provides the configuration forms for the FSC Safety Manager. It supports TPS Release 660 and the Honeywell FSC system Release 670. These forms should be used to assist you in configuring the FSC-SMM data points at the Universal Station.

The user should be familiar with the FSC Safety Manager system control functions described in *FSC Safety Manager Control Functions* and the *Implementation/Startup & Reconfiguration - 2* binder (LCN) bookset before using this publication.

**All references in this manual to "FSC Safety Manager" or "FSC Safety Manager Module" pertain only for use with the Honeywell FSC system.**



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# FSC Safety Manager Form Instructions

## FILLING OUT PAPER FORMS

### Where To Find Parameter and Data Point Help Information

Refer to the *FSC Safety Manager Parameter Reference Dictionary* and the *FSC Safety Manager Control Functions* publications in the *FSC Implementation Safety Manager* binder for further information on parameters and data points.

Please note that the sample form below is an example only and does not represent any existing configuration form.

FORMS →

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**FSC-SM Analog Input Data Point**

| FSC-SM Analog Input Data Point | Engineer: | Date: |
|--------------------------------|-----------|-------|
|--------------------------------|-----------|-------|

**NAME**  
Tag Name \_\_\_\_\_ *range = Not applicable  
default = Blank*

**PNTFORM**  
Point Form \_\_\_\_\_ *Point Assignment Display*

**Full** Point is fully displayed and alarmed  
**Componnt** Point is partially displayed but not alarmed

**PTDESC**  
Point Descriptor \_\_\_\_\_ *range = Not applicable  
default = Blank*

**EUDESC**  
Engineering Units Descriptor \_\_\_\_\_ *range = Not applicable  
default = Blank*

**KEYWORD**  
Point Keyword Descriptor \_\_\_\_\_ *range = Not applicable  
default = Blank*

**PVSRCOPT**  
PV Source Option **OnlyAuto** All *N/A if PNTFORM = Componnt.*  
PV source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available and Operator/CL program may supply PV.

**PVSOURCE**  
PV Source **Auto** Man Sub **Applies when PVSRCOPT = All.**  
Field wiring or memory fetch supplies PV. PV is supplied by Operator or program. A value is substituted by a CL program.

**Alarming display**

**PVHITP**  
PV High Alarm Trip Point \_\_\_\_\_ *range = PVLOTP to PVEUHI, & NaN  
default = NaN*

**PVHIPR**  
PV High Alarm Priority NoAction Journal **Low** High  
Emergency Printer JNLPrint

**PVLOTP**  
PV Low Alarm Trip Point \_\_\_\_\_ *range = PVEULO to PVHITP, & NaN  
default = NaN*

**PVLOPR**  
PV Low Alarm Priority NoAction Journal **Low** High  
Emergency Printer JNLPrint

**Universal Station's configuration display** where values and parameter options which have been written on the form will be keyed in and/or selected.

If parameters within a box do not apply when some other parameter option is selected, that box is indicated by **N/A if** or **ONLY if**.

Text in **bold** indicates what parameter(s) or parameter option(s) apply when an option is selected for a previous parameter.

Parameter values in **bold italics** indicate the default value for that parameter.

Some parameters will display only if a value has been specified for another parameter. For example, alarm priorities will not show until the corresponding alarm trip point has been specified. Thus, PVHIPR (PV High Alarm Priority) and PVLOPR (PV Low Alarm Priority) will not display until a value has been entered for the PVHITP (PV High Alarm Trip Point) and PVLOTP (PV Low Alarm Trip Point) parameters, respectively.

# FSC Safety Manager Form Instructions, continued

## ENTERING DATA FROM THE FORMS INTO THE SYSTEM

Select the **NETWORK INTERFACE MODULE** target from the Engineering Main Menu to get the NIM BUILD TYPE SELECT MENU.

|                                                |                                                            |                                                           |
|------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|
| FSC-SM NODE<br>CONFIGURATION<br>Form FS88-501A | FSC-SM<br>NODE-SPECIFIC<br>CONFIGURATION<br>Form FS88-501B | PROCESS<br>POINT<br>BUILDING<br>Form FS88-502 to FS88-510 |
|------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|



|                                                                   |            |         |       |      |       |         |
|-------------------------------------------------------------------|------------|---------|-------|------|-------|---------|
| <b>Select a point type</b> from this PROCESS POINT BUILDING menu. |            |         |       |      |       |         |
| Analg In/Out                                                      | Dig In/Out | DigComp | Logic | Flag | Timer | Numeric |

## FSC-SM UCN Node Configuration

|                               |           |       |
|-------------------------------|-----------|-------|
| FSC-SM UCN Node Configuration | Engineer: | Date: |
|-------------------------------|-----------|-------|

### UCN Node Configuration Display

#### NOTE

This form is intended primarily for use in configuring each FSC-SM on your system by selecting **SM** for **NODETYP** below. This same form can be used to configure each NIM on your system by selecting **NIM** for **NODETYP** (some different targets will appear as shown). If your system has other UCN devices (PM, APM, HPM, LM), the NIMs can also be configured using the forms contained in the documentation for those devices.

|                                                                                 |                                                         |                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NTWKNUM</b><br>NIM's UCN that SM is connected to<br>(Process Network Number) | <input type="text"/>                                    | <i>range = 1 to 20</i><br><i>default = Not applicable</i>                                                                                                                                                                 |
| <b>NODENUM</b><br>SMM's or NIM's Address on the UCN                             | <input type="text"/>                                    | <i>range = 1 to 64</i><br><i>default = Not applicable</i>                                                                                                                                                                 |
| <b>NODETYP</b><br>Type of Node on the UCN                                       | NotConfig<br>Nim<br>PM<br>APM<br>HPM<br>LM<br><b>SM</b> | This NIM or SM not configured on UCN<br>Network Interface Module<br>Process Manager Module<br>Advanced Process Manager Module<br>High Performance Process Manager Module<br>Logic Manager Module<br>Safety Manager Module |
| <b>SMPLTFM</b><br>Type of Safety Control Platform                               | <b>FSC</b><br>TRICON                                    | Applicable only if NODETYP = SM<br>Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller                                                                                                    |
| <b>NODEASSN</b>                                                                 | <b>ThisNim</b><br><br>RemotNim                          | NIM assigned to report alarms and perform<br>checkpointing for network specified in<br>NTWKNUM<br>Additional NIM on a network that already has<br>"ThisNIM" defined                                                       |
| <b>LOADSCOP</b><br>Load Scope                                                   | NimOnly<br><b>NimAndPm</b>                              | Applicable only if NODETYP = Nim<br>Configured data goes to the NIM only<br>Configured data goes to NIM and SM                                                                                                            |
| <b>NMSGTXT</b><br>Number of Message Text Items                                  | <input type="text"/>                                    | <i>range = 0 to 15</i><br><i>default = 0</i>                                                                                                                                                                              |
| <b>DIGITAL COMPOSITE PV STATE TEXT</b>                                          |                                                         |                                                                                                                                                                                                                           |
| <b>MOVPTXT</b><br>Moving Text                                                   | Moving                                                  | Individual point configuration specifies<br>which input state is associated with this moving<br>text. If NODENUM is even, this parameter is<br>hidden.                                                                    |

## FSC-SM **UCN Node** Configuration

*UCN Node Configuration Display (Continued)*

**BADPVTXT**  
Bad Text

Bad

Individual point configuration specifies which input state is associated with the bad text. If NODENUM is even, this parameter is hidden.

## FSC-SM UCN Node Configuration

|                               |           |       |
|-------------------------------|-----------|-------|
| FSC-SM UCN Node Configuration | Engineer: | Date: |
|-------------------------------|-----------|-------|

### Status Message Text Configuration Display

N/A if NMSGTXT = 0

|                                              |           |                                                                 |
|----------------------------------------------|-----------|-----------------------------------------------------------------|
| <b>MSGTXT(1)</b><br>Status Message Text #1   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(2)</b><br>Status Message Text #2   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(3)</b><br>Status Message Text #3   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(4)</b><br>Status Message Text #4   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(5)</b><br>Status Message Text #5   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(6)</b><br>Status Message Text #6   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(7)</b><br>Status Message Text #7   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(8)</b><br>Status Message Text #8   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(9)</b><br>Status Message Text #9   | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(10)</b><br>Status Message Text #10 | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(11)</b><br>Status Message Text #11 | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(12)</b><br>Status Message Text #12 | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(13)</b><br>Status Message Text #13 | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(14)</b><br>Status Message Text #14 | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |
| <b>MSGTXT(15)</b><br>Status Message Text #15 | . . . . . | <i>range</i> = Eight-Character String<br><i>default</i> = Blank |



## FSC-SM Node-Specific Configuration

|                      |           |       |
|----------------------|-----------|-------|
| FSC-SM Configuration | Engineer: | Date: |
|----------------------|-----------|-------|

**WARNING:** If the point mix or the scan rate is changed, then the node's full database will be set to its default.

### SM Node Configuration Display

|                                                      |                                     |                                                                                                                                                       |
|------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NTWKNUM</b><br>NIM's UCN that SMM is connected to | .                                   | range = 1 to 20<br>default = Not applicable                                                                                                           |
| <b>NODENUM</b><br>SMM's Address on the UCN           | .                                   | range = 1 to 64<br>default = Not applicable                                                                                                           |
| <b>NODETYP</b><br>Enumeration of the UCN Node Type   | PM<br>APM<br>HPM<br>LM<br><b>SM</b> | Process Manager Module<br>Advanced Process Manager Module<br>High Performance Process Manager Module<br>Logic Manager Module<br>Safety Manager Module |
| <b>SMPLTFM</b><br>Type of Safety Control Platform    | <b>FSC</b><br>TRICON                | Applicable only if NODETYP = SM<br>Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller                                |
| <b>NAISLOT*</b><br># of Analog Input Slots           | . . .                               | range = 0 to 1000<br>default = 0                                                                                                                      |
| <b>NAOSLOT*</b><br># of Analog Output Slots          | . . .                               | range = 0 to 1000<br>default = 0                                                                                                                      |
| <b>NDISLOT*</b><br># of Digital Input Slots          | . . .                               | range = 0 to 2000<br>default = 0                                                                                                                      |
| <b>NDOSLOT*</b><br># of Digital Output Slots         | . . .                               | range = 0 to 2000<br>default = 0                                                                                                                      |
| <b>NLOGSLOT*</b><br># of Logic Slots                 | .                                   | range = 0 to 30<br>default = 0                                                                                                                        |
| <b>NDCSLOT*</b><br># of Digital Composite Slots      | . . .                               | range = 0 to 1000                                                                                                                                     |
| <b>NONETXT</b><br>Digital Composite None State       | . . . . .                           | range = Not applicable<br>default = None                                                                                                              |
| <b>NNUMERIC**</b><br># of Numerics                   | . . .                               | range = 0 to 1000<br>default = 0                                                                                                                      |

\* The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired. The FSC-SMM will perform Range checking. The 0.5 second scan rates will tend to lower these maximum range values.

\*\* The actual Range check for this Parameter will depend on the NNLSBA. The FSC-SMM will perform Range checking.

## FSC-SM Node-Specific Configuration

### SM Node Configuration Display (continued)

**NNLSBA\*\*\***

FSC-SM Start Address of the Numeric Array

range = 0 to 65535

default = 0 (Not Configured)

**NFLAG**

# of Flags

range = 0 to 2000

default = 0

**The Range check for this Parameter will be dependent on the FLLSBA. SMM will perform Range checking.****FLLSBA**

FSC-SM Start Address of the Flag Array

range = 0 to 65535

default = 0 (Not Configured)

**The Range check for this Parameter will be dependent on the NFLAG value. SMM will perform Range checking.****NTIMER**

# of Timers

range = 0 to 1500

default = 0

**The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired. The 0.5 second scan rates will tend to lower the maximum range value. SMM will perform Range checking.****SCANRATE**

SMM's Scan Rate in Times per Second

range: 0 = Null

1 = AR1DT2

2 = AR2DT2

3 = AR1DT1

default = AR2DT2

**\*\*\* The Range check for this Parameter will be dependent on NNUMERIC value. The FSC-SMM will perform Range checking.**

## FSC-SM Analog Input Data Point

|                                |           |       |
|--------------------------------|-----------|-------|
| FSC-SM Analog Input Data Point | Engineer: | Date: |
|--------------------------------|-----------|-------|

### Point Assignment Display

|                                                      |                                     |                                                                                                                   |
|------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>NAME</b><br>Tag Name                              | <input type="text"/>                | <i>range</i> = Not applicable<br><i>default</i> = Blank                                                           |
| <b>NODETYP</b><br>Node Type                          | PM<br>APM<br>HPM<br>LM<br><b>SM</b> | <i>range</i> = Not applicable<br><i>default</i> = PM                                                              |
| <b>SMPLTFM</b><br>Type of Safety Control Platform    | <b>FSC</b><br>TRICON                | Applicable if NODETYP = SM<br>Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller |
| <b>PNTFORM</b><br>Point Form                         | <b>Full</b><br>Componnt             | Point is fully displayed and alarmed<br>Point is partially displayed but not alarmed                              |
| <b>PTDESC</b><br>Point Descriptor                    | <input type="text"/>                | <i>range</i> = Not applicable<br><i>default</i> = Blank                                                           |
| <b>EUDESC</b><br>Engineering Unit Descriptor         | <input type="text"/>                | <i>range</i> = Not applicable<br><i>default</i> = Blank                                                           |
| <b>KEYWORD</b><br>Point Keyword Descriptor           | <input type="text"/>                | <i>range</i> = Not applicable<br><i>default</i> = Blank                                                           |
| <b>ASSOCDSP</b><br>Associated Display                | <input type="text"/>                | <i>range</i> = Alphanumeric<br><i>default</i> = Blank                                                             |
| <b>UNIT</b><br>Process Unit Point is Assigned to     | <input type="text"/>                | <i>range</i> = Alphanumeric<br><i>default</i> = Not applicable                                                    |
| <b>NTWKNUM</b><br>NIM's UCN that Contains This Point | <input type="text"/>                | <i>range</i> = 1 to 20<br><i>default</i> = Not applicable                                                         |
| <b>NODENUM</b><br>SMM's Address on the UCN           | <input type="text"/>                | <i>range</i> = 1 to 64<br><i>default</i> = Not applicable                                                         |
| <b>SLOTNUM</b><br>Point Slot Number                  | <input type="text"/>                | <i>range</i> = 1 to 1000<br><i>default</i> = Not applicable                                                       |

# FSC-SM Analog Input Data Point

Point Assignment Display (continued)

|                                                      |                                                          |                                                                  |                                                                                                                                               |
|------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      |                                                          | N/A if PNTFORM = Component                                       |                                                                                                                                               |
| <b>PRIMMOD</b><br>Primary Module Point ID            | <input type="text"/>                                     |                                                                  | <i>range</i> = Not applicable<br><i>default</i> = Null                                                                                        |
| <b>PLCADDR</b><br>FSC Source Address                 | <input type="text"/>                                     | <i>range</i> = 0 to 65535<br><i>default</i> = 0 (Not Configured) |                                                                                                                                               |
| <i>NIM/PV Configuration Display</i>                  |                                                          |                                                                  |                                                                                                                                               |
| <b>PVCHAR</b><br>PV Characterization                 | <div> <div><b>LINEAR</b></div> <div>SQROOT</div> </div>  |                                                                  |                                                                                                                                               |
| <b>PVRAWHI</b><br>PV Raw High Value                  | <input type="text"/>                                     | <i>range</i> = > PVRAWLO<br><i>default</i> = 4095                |                                                                                                                                               |
| <b>PVRAWLO</b><br>PV Raw Low Value                   | <input type="text"/>                                     | <i>range</i> = < PVRAWHI<br><i>default</i> = 0                   |                                                                                                                                               |
| <b>INPTDIR</b><br>Analog Input Direction             | <div> <div><b>Direct</b></div> <div>Reverse</div> </div> |                                                                  | Highest energy from sensor = 100% PV<br>Highest energy from sensor = 0% PV                                                                    |
| <b>PVEUHI</b><br>PV High Range in EUs                | <input type="text"/>                                     | <i>range</i> = PVEULO to PVEXEUHI<br><i>default</i> = 100.0      |                                                                                                                                               |
| <b>PVEULO</b><br>PV Low Range in EUs                 | <input type="text"/>                                     | <i>range</i> = PVEXEULO to PVEUHI<br><i>default</i> = 0.0        |                                                                                                                                               |
| <b>PVFORMAT</b><br>PV Decimal Point Format           | <div> <div>D0</div> <div><b>D1</b></div> </div>          | <div> <div>D2</div> <div>D3</div> </div>                         | <div> <div>D0 = 9999<b>o</b></div> <div>D1 = 999<b>o</b>9</div> </div> <div> <div>D2 = 99<b>o</b>99</div> <div>D3 = 9<b>o</b>999</div> </div> |
| <b>PVEXEUHI</b><br>PV Range Extension High in EUs    | <input type="text"/>                                     | <i>range</i> = ≥ PVEUHI<br><i>default</i> = 102.9                |                                                                                                                                               |
| <b>PVEXEULO</b><br>PV Range Extension Low in EUs     | <input type="text"/>                                     | <i>range</i> = < PVEULO<br><i>default</i> = -2.9                 |                                                                                                                                               |
| <b>PVCLAMP</b><br>PV Clamping Option                 | <div> <div><b>NoClamp</b></div> <div>Clamp</div> </div>  |                                                                  | No clamping of the PV value<br>Clamp PV value at range extension limit                                                                        |
| <b>LOCUTOFF</b><br>Low Signal Cutoff for Flow Inputs | <input type="text"/>                                     | <i>range</i> = PVEULO to PVEUHI and NaN<br><i>default</i> = NaN  |                                                                                                                                               |

## FSC-SM Analog Input Data Point

*NIM/PV Configuration Display (continued)*

|                                            |                            |                                                                                                                                                                       |
|--------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            |                            | N/A if PNTFORM = Componnt                                                                                                                                             |
| <b>PVSRCOPT</b><br>PV Source Option        | <b>OnlyAuto</b><br><br>All | PV source selection is not available and field wiring or memory fetch supplies PV.<br>Selection from PVSOURCE is available and Operator/CL-program may supply PV.     |
| <b>PVSOURCE</b><br>PV Source               | <b>Auto</b><br>Man<br>Sub  | <b>Applies when PVSRCOPT = All.</b><br>Field wiring or memory fetch supplies PV.<br>PV is supplied by Operator or program.<br>A value is substituted by a CL program. |
| <b>TF</b><br>PV Filter Lag Time in Minutes | . . .                      | <i>range</i> = 0.0 to 60.0 (minutes)<br><i>default</i> = 0.0 (PV will not be delayed)                                                                                 |
| <b>PVTV</b><br>PV Target Value in EUs      | . . . . .                  | N/A if PNTFORM = Componnt<br><i>range</i> = PVEXEULO to PVEXEUHI, and NaN<br><i>default</i> = NaN                                                                     |
| <b>OVERVAL</b><br>Overview Display Value   | . . .                      | <i>range</i> = 0 to 100<br><i>default</i> = 25                                                                                                                        |

# FSC-SM Analog Input Data Point

NIM/PV Configuration Display (continued)

## Alarming Display

N/A if PNTFORM = Componnt

The following two example parameters represent the typical format of alarm trip points and priorities. Please refer to these as you are entering trip points and alarm priorities that follow this information. **NOTE:** If you do not enter a trip point value for a given pair of trip points and alarm priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

|                                                           |                                                     |                                                                                                                                                                                                                                     |
|-----------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>\$AUXUNIT</b><br>Auxiliary Unit                        | _ _                                                 | <i>range</i> = Any configured unit in NCF<br><i>default</i> = - -                                                                                                                                                                   |
| <b>\$ALDLOTP</b><br>Alarm Delay Option                    |                                                     | <i>range</i> = None, OffDlyLo, OffDlyHi, OffDlyEm, OnDlyLo, OnDlyHi, OnDlyEm                                                                                                                                                        |
| <b>DLYTIME</b><br>Alarm delay in seconds                  |                                                     | Applies when \$ALDLOPT other than NONE<br><i>range</i> = 0 to 60<br><i>default</i> = 5 (seconds)                                                                                                                                    |
| <b>\$REALARM</b><br>Re-alarming Interval (R660 and later) | _ _ _                                               | <i>range</i> = 0 to 9999 minutes<br><i>default</i> = 0 minutes                                                                                                                                                                      |
| <b>PVALDB</b><br>PV Alarm Deadband in %                   | Half (of 1%)<br>Four                                | N/A if PNTFORM = Componnt<br><b>One</b> Two Three<br>Five EU                                                                                                                                                                        |
| <b>PVHITP</b><br>PV High Alarm Trip Point                 | _ _ _ _ _ _ _                                       | <i>range</i> = PVLOTP to PVEUHI, and NaN<br><i>default</i> = NaN                                                                                                                                                                    |
| <b>PVLOTP</b><br>PV Low Alarm Trip Point                  | _ _ _ _ _ _ _                                       | <i>range</i> = PVEULO to PVHITP, and NaN<br><i>default</i> = NaN                                                                                                                                                                    |
| <b>BADPVPR</b><br>BAD PV Alarm Priority                   | <b>NoAction</b><br>Journal<br>Low High<br>Emergency | Alarm is not reported to the system (R670 and later)<br>Logged but not reported to Operator Stations<br>Reported to Unit Alarm Summary display<br>Reported to Area and Unit Alarm Summary<br>Reported to all Alarm Summary displays |

# FSC-SM Analog Input Data Point

NIM/PV Configuration Display (continued)

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## Alarming Display

N/A if PNTFORM = Componnt, Only if PVALDB = EU

|                                            |           |                                                 |
|--------------------------------------------|-----------|-------------------------------------------------|
| <b>PVALDBEU</b><br>PV Alarm Deadband in EU | . . . . . | <i>range</i> = $\geq 0$<br><i>default</i> = 1.0 |
|--------------------------------------------|-----------|-------------------------------------------------|

N/A if PNTFORM = Componnt

|                                                                        |                                     |                    |                                                                                                    |                                                                                 |
|------------------------------------------------------------------------|-------------------------------------|--------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>PVHIPR</b><br>PV High Alarm Priority                                | <b>NoAction</b> (R670)<br>Emergency | Journal<br>Printer | Low<br>Jnlprint                                                                                    | High                                                                            |
| <b>PVLOPR</b><br>PV Low Alarm Priority                                 | <b>NoAction</b> (R670)<br>Emergency | Journal<br>Printer | Low<br>Jnlprint                                                                                    | High                                                                            |
| <b>ALENBST</b><br>Point Alarm Enable Status                            | <b>Enable</b><br>Disable<br>Inhibit |                    | <u>Displayed</u><br>Yes<br>No<br>No                                                                | <u>Logged</u><br>Yes<br>Yes<br>No<br><u>Reported to EIP</u><br>Yes<br>Yes<br>No |
| <b>PRIMMOD</b><br>Primary Module ID                                    | . . . . .                           |                    | <i>range</i> = Not applicable<br><i>default</i> = Null                                             |                                                                                 |
| <b>CCSRC</b><br>Target Address in FSC for the<br>Contact Cutout Source | . . . . .                           |                    | <i>range</i> = 0 to 65535<br><i>default</i> = 0 (Not configured/UCN Access allowed for<br>CONTCUT) |                                                                                 |



## FSC-SM Analog Output Data Point

|                             |           |       |
|-----------------------------|-----------|-------|
| FSC-SM AnalogOut Data Point | Engineer: | Date: |
|-----------------------------|-----------|-------|

### Point Assignment Display

|                |           |                               |
|----------------|-----------|-------------------------------|
| <b>NAME</b>    |           | <i>range =</i> Not applicable |
| Tag Name       | . . . . . | <i>default =</i> Blank        |
| <b>NODETYP</b> | PM        | <i>range =</i> Not applicable |
| Node Type      | APM       | <i>default =</i> PM           |
|                | HPM       |                               |
|                | LM        |                               |
|                | <b>SM</b> |                               |

Applicable if NODETYP = SM

|                                    |             |                                                                                     |
|------------------------------------|-------------|-------------------------------------------------------------------------------------|
| <b>SMPLTFM</b>                     | <b>FSC</b>  |                                                                                     |
| Type of Safety Control Platform    | TRICON      | Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller |
| <b>PNTFORM</b>                     | <b>Full</b> | Point is fully displayed and alarmed                                                |
| Point Form                         | Componnt    | Point is partially displayed but not alarmed                                        |
| <b>PTDESC</b>                      |             | <i>range =</i> Not applicable                                                       |
| Point Descriptor                   | . . . . .   | <i>default =</i> Blank                                                              |
| <b>EUDESC</b>                      | . . . . .   | <i>range =</i> Not applicable                                                       |
| Engineering Units Descriptor       |             | <i>default =</i> Blank                                                              |
| <b>KEYWORD</b>                     | . . . . .   | <i>range =</i> Not applicable                                                       |
| Keyword Descriptor                 |             | <i>default =</i> Blank                                                              |
| <b>ASSOCDSP</b>                    | . . . . .   | <i>range =</i> Alphanumeric                                                         |
| Associated Display                 |             | <i>default =</i> Blank                                                              |
| <b>UNIT</b>                        | . .         | <i>range =</i> Alphanumeric                                                         |
| Process Unit Point is Assigned to  |             | <i>default =</i> Not applicable                                                     |
| <b>NTWKNUM</b>                     | . .         | <i>range =</i> 1 to 20                                                              |
| NIM's UCN that Contains This Point |             | <i>default =</i> Not applicable                                                     |
| <b>NODENUM</b>                     | . .         | <i>range =</i> 1 to 64                                                              |
| SMM's Address on the UCN           |             | <i>default =</i> Not applicable                                                     |
| <b>SLOTNUM</b>                     | . . .       | <i>range =</i> 1 to 1000                                                            |
| Point Slot Number                  |             | <i>default =</i> Not applicable                                                     |

The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.

## FSC-SM Analog Output Data Point

### Point Assignment Display (continued)

|                         |           |                                     |
|-------------------------|-----------|-------------------------------------|
|                         |           | N/A if PNTFORM = Component          |
| <b>PRIMMOD</b>          |           | <i>range</i> = Not applicable       |
| Primary Module Point ID |           | <i>default</i> = Null               |
|                         | . . . . . |                                     |
| <b>PLCADDR</b>          | . . . . . | <i>range</i> = 0 to 65535           |
| FSC Source Address      |           | <i>default</i> = 0 (Not Configured) |
| <b>OPFINHI</b>          | . . . . . | <i>range</i> = > OPFINLO            |
|                         |           | <i>default</i> = 4095.0             |
| <b>OPFINLO</b>          | . . . . . | <i>range</i> = < OPFINHI            |
|                         |           | <i>default</i> = 0.0                |

### Operating Configuration Display

|                          |               |                                                 |
|--------------------------|---------------|-------------------------------------------------|
| <b>OPTDIR</b>            | <b>Direct</b> | For final OP: 100% = Highest energy to actuator |
| AO Direct/Reverse Action | Reverse       | For final OP: 100% = Lowest energy to actuator  |

### Mode Configuration Display

|                                   |             |                                                                        |
|-----------------------------------|-------------|------------------------------------------------------------------------|
|                                   |             | N/A if PNTFORM = Component                                             |
| <b>RCASOPT</b>                    | <b>None</b> | No cascade mode of <u>any type</u> is allowed                          |
| Remote Cascade Option             | Ddc         | In Cas, AM point controls this point's OP                              |
| <b>NMODE</b>                      |             | <b>None, Man applies when RCASOPT = None.</b>                          |
| Analog Output Point's Normal Mode | <b>None</b> | <b>None, Man, Cas applies when RCASOPT = Ddc.</b>                      |
|                                   | Man         | No configured "normal" operating mode.                                 |
|                                   | Cas         | Manual is the configured "normal" mode.                                |
|                                   |             | If RCASOPT equals Ddc, Cascade may be configured as the "normal" mode. |

|                       |               |                                                |
|-----------------------|---------------|------------------------------------------------|
|                       |               | N/A if NMODE = None, or if PNTFORM = Component |
| <b>NMODATTR</b>       | <b>None</b>   | Parameter NMODE sets "normal" mode             |
| Normal Mode Attribute | Operator      | Operator sets "normal" operating mode          |
|                       | Program       | Program sets "normal" operating mode           |
| <b>MODEPERM</b>       | <b>Permit</b> | Operator can change this point's mode          |
| Mode Permissive       | NotPerm       | Operator cannot change this point's mode       |

## FSC-SM Digital Composite Data Point

|                           |           |       |
|---------------------------|-----------|-------|
| FSC-SM DigComp Data Point | Engineer: | Date: |
|---------------------------|-----------|-------|

### Point Assignment Display

|                             |                                     |                                                         |
|-----------------------------|-------------------------------------|---------------------------------------------------------|
| <b>NAME</b><br>Tag Name     | <input type="text"/>                | <i>range =</i> Not applicable<br><i>default =</i> Blank |
| <b>NODETYP</b><br>Node Type | PM<br>APM<br>HPM<br>LM<br><b>SM</b> | <i>range =</i> Not applicable<br><i>default =</i> PM    |

Applicable if NODETYP = SM

|                                                      |                         |                                                                                      |
|------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------|
| <b>SMPLTFM</b><br>Type of Safety Control Platform    | <b>FSC</b><br>TRICON    | Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller  |
| <b>PNTFORM</b><br>Point Form                         | <b>Full</b><br>Componnt | Point is fully displayed and alarmed<br>Point is partially displayed but not alarmed |
| <b>PTDESC</b><br>Point Descriptor                    | <input type="text"/>    | <i>range =</i> Not applicable<br><i>default =</i> Blank                              |
| <b>EUDESC</b><br>Engineering Units Descriptor        | <input type="text"/>    | <i>range =</i> Not applicable<br><i>default =</i> Blank                              |
| <b>KEYWORD</b><br>Keyword Descriptor                 | <input type="text"/>    | <i>range =</i> Not applicable<br><i>default =</i> Blank                              |
| <b>ASSOCDSP</b><br>Associated Display                | <input type="text"/>    | <i>range =</i> Alphanumeric<br><i>default =</i> Blank                                |
| <b>UNIT</b><br>Process Unit Point is Assigned to     | <input type="text"/>    | <i>range =</i> Alphanumeric<br><i>default =</i> Not applicable                       |
| <b>NTWKNUM</b><br>NIM's UCN that Contains This Point | <input type="text"/>    | <i>range =</i> 1 to 20<br><i>default =</i> Not applicable                            |
| <b>NODENUM</b><br>SMM's Address on the UCN           | <input type="text"/>    | <i>range =</i> 1 to 64<br><i>default =</i> Not applicable                            |
| <b>SLOTNUM</b><br>Point Slot Number                  | <input type="text"/>    | <i>range =</i> 1 to 1000<br><i>default =</i> Not applicable                          |

The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.

N/A if PNTFORM = Componnt

|                                           |                      |                                                        |
|-------------------------------------------|----------------------|--------------------------------------------------------|
| <b>PRIMMOD</b><br>Primary Module Point ID | <input type="text"/> | <i>range =</i> Not applicable<br><i>default =</i> Null |
|-------------------------------------------|----------------------|--------------------------------------------------------|

## FSC-SM Digital Composite Data Point

*Point Assignment Display (continued)*

|                                      |          |                                           |
|--------------------------------------|----------|-------------------------------------------|
| <b>NOSTATES</b>                      | <b>2</b> | 2 digital switch states can be configured |
| Number of Digital States             | 3        | 3 digital switch states can be configured |
| <b>NODINPTS</b>                      | <b>0</b> | 0 digin connections can be configured     |
| Number of Digital Input Connections  | <b>1</b> | 1 digin connection can be configured      |
|                                      | 2        | 2 digin connections can be configured     |
| <b>NODOPTS</b>                       | <b>0</b> | 0 digout connections can be configured    |
| Number of Digital Output Connections | <b>1</b> | 1 digout connection can be configured     |
|                                      | 2        | 2 digout connections can be configured    |
|                                      | 3        | 3 digout connections can be configured    |

### State Configuration Display

N/A if PNTFORM = Componnt

|                     |                                |                                    |       |         |                                                                               |
|---------------------|--------------------------------|------------------------------------|-------|---------|-------------------------------------------------------------------------------|
|                     |                                |                                    |       |         | (1) = First active state<br>(0) = Inactive state<br>(2) = Second active state |
| <b>STATETXT (1)</b> |                                |                                    |       |         |                                                                               |
| State 1 Descriptor  | (1)                            | . . . . .                          |       |         | <i>range</i> = Not applicable<br><i>default</i> = On                          |
| <b>STATETXT (0)</b> |                                |                                    |       |         |                                                                               |
| State 0 Descriptor  | (0)                            | . . . . .                          |       |         | <i>range</i> = Not applicable<br><i>default</i> = Off                         |
| <b>STATETXT (2)</b> |                                |                                    |       |         |                                                                               |
| State 2 Descriptor  | (2)                            | . . . . .                          |       |         | <i>range</i> = Not applicable<br><i>default</i> = "State2"                    |
| <b>BOXCLR (1)</b>   | Red                            | <b>Green</b> (Middle box default)  | White | Black   |                                                                               |
| State 1 Box Color   | Cyan                           | Yellow                             | Blue  | Magenta |                                                                               |
| <b>BOXCLR (0)</b>   | Red                            | Green                              | White | Black   |                                                                               |
| State 0 Box Color   | Cyan                           | <b>Yellow</b> (Middle box default) | Blue  | Magenta |                                                                               |
| <b>BOXCLR (2)</b>   | <b>Red</b> (Lower box default) | Green                              | White | Black   |                                                                               |
| State 2 Box Color   | Cyan                           | Yellow                             | Blue  | Magenta |                                                                               |

A horizontal number line is shown, starting at 0 and ending at 100. There are major tick marks at intervals of 10, labeled 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100. A single black dot is placed on the tick mark for 70.

# FSC-SM Digital Composite Data Point

## PV/Input Configuration Display

Applies if NODINPTS = 1

|                 |                 |                                       |
|-----------------|-----------------|---------------------------------------|
| <b>D1_1</b>     | <b>PVState1</b> | Use and display contents of STATETXT1 |
| D1_One PV State | PVState0        | Use and display contents of STATETXT0 |

Applies if NODINPTS = 2

|                         |                 |                                       |
|-------------------------|-----------------|---------------------------------------|
| <b>D2D1_00</b>          | PVState1        | Use and display contents of STATETXT1 |
| D2D1 Zero_Zero PV State | PVState0        | Use and display contents of STATETXT0 |
|                         | PVState2        | Use and display contents of STATETXT2 |
|                         | BadPV           | Use and display contents of BADPVTXT  |
|                         | <b>MovPV</b>    | Use and display contents of MOVPVTXT  |
| <b>D2D1_01</b>          | <b>PVState1</b> | Use and display contents of STATETXT1 |
| D2D1 Zero_One PV State  | PVState0        | Use and display contents of STATETXT0 |
|                         | PVState2        | Use and display contents of STATETXT2 |
|                         | BadPV           | Use and display contents of BADPVTXT  |
|                         | MovPV           | Use and display contents of MOVPVTXT  |
| <b>D2D1_10</b>          | PVState1        | Use and display contents of STATETXT1 |
| D2D1 One_Zero PV State  | <b>PVState0</b> | Use and display contents of STATETXT0 |
|                         | PVState2        | Use and display contents of STATETXT2 |
|                         | BadPV           | Use and display contents of BADPVTXT  |
|                         | MovPV           | Use and display contents of MOVPVTXT  |
| <b>D2D1_11</b>          | PVState1        | Use and display contents of STATETXT1 |
| D2D1 One_One PV State   | PVState0        | Use and display contents of STATETXT0 |
|                         | PVState2        | Use and display contents of STATETXT2 |
|                         | MovPV           | Use and display contents of MOVPVTXT  |
|                         | <b>BadPV</b>    | Use and display contents of BADPVTXT  |

N/A if PNTFORM = Componnt, or if NODINPTS = 0

|                  |             |                                                                                    |
|------------------|-------------|------------------------------------------------------------------------------------|
| <b>PVSRCOPT</b>  | OnlyAuto    | PV source selection is not available and field wiring or memory fetch supplies PV. |
| PV Source Option | <b>All</b>  | Selection from PVSOURCE is available and Operator/CL-program may supply PV.        |
| <b>PVSOURCE</b>  |             | <b>Applies when PVSRCOPT = All.</b>                                                |
| PV Source        | <b>Auto</b> | Field wiring or memory fetch supplies PV.                                          |
|                  | Man         | PV is supplied by Operator or program.                                             |
|                  | Sub         | A value is substituted by a CL-program.                                            |
|                  | Track       | PV tracks the output.                                                              |

# FSC-SM Digital Composite Data Point

Output Configuration Display

N/A if NODOPTS < 2

|                                             |                  |
|---------------------------------------------|------------------|
| <b>NONECONF</b><br>None State Configuration | On<br><b>Off</b> |
|---------------------------------------------|------------------|

N/A if NODOPTS = 0

| <b>STATE 1</b>                                     | <u>OUTPUT 1</u>  | <u>OUTPUT 2</u>  | <u>OUTPUT 3</u>  |
|----------------------------------------------------|------------------|------------------|------------------|
| <b>ST1_OP1</b> (NODOPTS = 1)<br>State 1 – Output 1 | <b>Off</b><br>On | Not applicable   | Not applicable   |
| <b>ST1_OP2</b> (NODOPTS = 2)<br>State 1 – Output 2 | <b>Off</b><br>On | <b>Off</b><br>On | Not applicable   |
| <b>ST1_OP3</b> (NODOPTS = 3)<br>State 1 – Output 3 | <b>Off</b><br>On | <b>Off</b><br>On | <b>Off</b><br>On |
| <b>STATE 0</b>                                     |                  |                  |                  |
| <b>ST0_OP1</b> (NODOPTS = 1)<br>State 0 – Output 1 | <b>Off</b><br>On | Not applicable   | Not applicable   |
| <b>ST0_OP2</b> (NODOPTS = 2)<br>State 0 – Output 2 | <b>Off</b><br>On | <b>Off</b><br>On | Not applicable   |
| <b>ST0_OP3</b> (NODOPTS = 3)<br>State 0 – Output 3 | <b>Off</b><br>On | <b>Off</b><br>On | <b>Off</b><br>On |
| <b>STATE 2</b>                                     |                  |                  |                  |
| <b>ST2_OP1</b> (NODOPTS = 1)<br>State 2 – Output 1 | <b>Off</b><br>On | Not applicable   | Not applicable   |
| <b>ST2_OP2</b> (NODOPTS = 2)<br>State 2 – Output 2 | <b>Off</b><br>On | <b>Off</b><br>On | Not applicable   |
| <b>ST2_OP3</b> (NODOPTS = 3)<br>State 2 – Output 3 | <b>Off</b><br>On | <b>Off</b><br>On | <b>Off</b><br>On |

N/A if NONECONF = OFF

| <b>NONE STATE</b>                                      | <u>OUTPUT 1</u>  | <u>OUTPUT 2</u>  | <u>OUTPUT 3</u>  |
|--------------------------------------------------------|------------------|------------------|------------------|
| <b>NONE_OP1</b> (NODOPTS = 1)<br>None State – Output 1 | <b>Off</b><br>On | Not applicable   | Not applicable   |
| <b>NONE_OP2</b> (NODOPTS = 2)<br>None State – Output 2 | <b>Off</b><br>On | <b>Off</b><br>On | Not applicable   |
| <b>NONE_OP3</b> (NODOPTS = 3)<br>None State – Output 3 | <b>Off</b><br>On | <b>Off</b><br>On | <b>Off</b><br>On |

# FSC-SM Digital Composite Data Point

Output Configuration Display (continued)

N/A if NODOPTS = 0

## DODSTN

Specification if the Digital Composite Input point. The destination can be specified by TAG-PARAMETER, the destination address in FSC.

### Output

### Source Point.Parameter

|            |                       |
|------------|-----------------------|
| DODSTN (1) | . . . . .   . . . . . |
| DODSTN (2) | . . . . .   . . . . . |
| DODSTN (3) | . . . . .   . . . . . |

## PISRC

|                                                           |           |                              |
|-----------------------------------------------------------|-----------|------------------------------|
| PISRC (1)                                                 | . . . . . | range = 0 to 65535           |
| Target Addresses in FSC for Permissive Interlock Source 1 |           | default = 0 (Not Configured) |

|                                                           |           |                              |
|-----------------------------------------------------------|-----------|------------------------------|
| PISRC (2)                                                 | . . . . . | range = 0 to 65535           |
| Target Addresses in FSC for Permissive Interlock Source 2 |           | default = 0 (Not Configured) |

|                                                           |           |                              |
|-----------------------------------------------------------|-----------|------------------------------|
| PISRC (3)                                                 | . . . . . | range = 0 to 65535           |
| Target Addresses in FSC for Permissive Interlock Source 3 |           | default = 0 (Not Configured) |

## OISRC

|                                                         |           |                              |
|---------------------------------------------------------|-----------|------------------------------|
| OISRC (1)                                               | . . . . . | range = 0 to 65535           |
| Target Addresses in FSC for Override Interlock Source 1 |           | default = 0 (Not Configured) |

|                                                         |           |                              |
|---------------------------------------------------------|-----------|------------------------------|
| OISRC (2)                                               | . . . . . | range = 0 to 65535           |
| Target Addresses in FSC for Override Interlock Source 2 |           | default = 0 (Not Configured) |

|                                                         |           |                              |
|---------------------------------------------------------|-----------|------------------------------|
| OISRC (3)                                               | . . . . . | range = 0 to 65535           |
| Target Addresses in FSC for Override Interlock Source 3 |           | default = 0 (Not Configured) |

## LMSRC

|                                                 |           |                              |
|-------------------------------------------------|-----------|------------------------------|
| LMSRC                                           | . . . . . | range = 0 to 65535           |
| Target Addresses in FSC for Local/Manual Source |           | default = 0 (Not Configured) |

# FSC-SM Digital Composite Data Point

## Mode Configuration Display

N/A if NODOPTS = 0

|                                          |                                    |                                                                                                                     |
|------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>NMODATTR</b><br>Normal Mode Attribute | <b>None</b><br>Operator<br>Program | Parameter NMODE sets "normal" mode<br>Operator sets "normal" operating mode<br>Program sets "normal" operating mode |
| <b>MODEPERM</b><br>Mode Permissive       | <b>Permit</b><br>NotPerm           | Operator can change this point's mode attribute<br>Operator cannot change this point's mode attribute               |
| <b>OROPT</b><br>Interlock Bypass Option  | <b>Off</b><br>On                   |                                                                                                                     |
| <b>BYPASS</b><br>Interlock Bypass State  | <b>Off</b><br>On                   | <b>Applies when OROPT = On.</b>                                                                                     |

## Alarming Display

N/A if PNTFORM = Componnt, or if NODINPTS = 0

|                                                         |                                       |                                                                                                                                                                                                                                           |                  |               |                        |     |     |     |    |     |     |    |    |    |
|---------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------|------------------------|-----|-----|-----|----|-----|-----|----|----|----|
| <b>\$AUXUNIT</b><br>Auxiliary Unit                      | .                                     | range = Any configured unit in NCF<br>default = - -                                                                                                                                                                                       |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>\$REALARM</b><br>Re-alarms Interval (R660 and later) | _ _ _ .                               | range = 0 to 9999 minutes<br>default = 0 minutes                                                                                                                                                                                          |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>ALMOPT</b><br>Alarming Option                        | <b>None</b><br>Cmddis<br><br>Offnorml | No alarms<br>Alarm if field device did not respond within configured feedback time duration.<br>Alarm if PV and PVNORMAL differs.                                                                                                         |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>FBTIME</b><br>Feedback Time in Seconds               | . . .                                 | <b>Applies when ALMOPT = Cmddis.</b><br>range = 1 to 1000<br>default = 1000                                                                                                                                                               |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>CMDFALTM</b><br>Command Failure Timeout              | . . .                                 | <b>Applies when ALMOPT = Cmddis.</b><br>range = < FBTIME<br>default = 0                                                                                                                                                                   |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>PVNORMAL</b><br>PV Value's Normal State Text         | <b>Off</b>                            | <b>Applies when ALMOPT = Offnorml.</b><br>range = STATETXT(0) – STATETXT(2)                                                                                                                                                               |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>ALENBST</b><br>Point Alarm Enable Status             |                                       | <b>Applies when ALMOT = Offnorml, Cmddis.</b>                                                                                                                                                                                             |                  |               |                        |     |     |     |    |     |     |    |    |    |
|                                                         | <b>Enable</b><br>Disable<br>Inhibit   | <table><tr><td><u>Displayed</u></td><td><u>Logged</u></td><td><u>Reported to EIP</u></td></tr><tr><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>No</td><td>Yes</td><td>Yes</td></tr><tr><td>No</td><td>No</td><td>No</td></tr></table> | <u>Displayed</u> | <u>Logged</u> | <u>Reported to EIP</u> | Yes | Yes | Yes | No | Yes | Yes | No | No | No |
| <u>Displayed</u>                                        | <u>Logged</u>                         | <u>Reported to EIP</u>                                                                                                                                                                                                                    |                  |               |                        |     |     |     |    |     |     |    |    |    |
| Yes                                                     | Yes                                   | Yes                                                                                                                                                                                                                                       |                  |               |                        |     |     |     |    |     |     |    |    |    |
| No                                                      | Yes                                   | Yes                                                                                                                                                                                                                                       |                  |               |                        |     |     |     |    |     |     |    |    |    |
| No                                                      | No                                    | No                                                                                                                                                                                                                                        |                  |               |                        |     |     |     |    |     |     |    |    |    |

## FSC-SM Digital Composite Data Point

*Alarming Display (continued)*

N/A if NODOPTS = 0

|                                                                         |                                                                               |                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OVRDALOP</b><br>Override Alarm Option                                | <b>None</b><br>Auto_Rtn<br>Cnfm_Rqd                                           | No alarms are generated.<br>Alarms return when override clears.<br>Alarms must be confirmed.                                                                                                                                                                                                                                                  |
| <b>BADPVPR</b><br><b>BADCTLPR</b><br><b>CMDDISPR</b><br><b>OFFNRMPR</b> | <b>NoAction</b><br>Journal<br>Low<br>High<br>Emergency<br>Printer<br>Jnlprint | <b>Applies when ALMOPT = Offnorml, Cmddis.</b><br>Alarm is not reported to the system. (R670 and later)<br>Logged but not reported to Op Stations.<br>Reported to Unit Alarm Summary display<br>Reported to Area and Unit Alarm Summary<br>Reported to all Alarm Summary displays<br>Reported to US RTJ only<br>Reported to US RTJ and logged |
| <b>EVTOPT</b><br>Event Report Option                                    | <b>None</b><br>Eip                                                            | <b>Applies when ALMOPT = None, Offnorml, Cmddis.</b><br>Neither Eip or Soe is allowed.<br>Process special is triggered in AM/CG.                                                                                                                                                                                                              |
| <b>EIPPCODE</b><br>Event Init. Proc. Destination<br>  . . . . .         |                                                                               | <b>Applies when EVTOPT = Eip.</b><br><i>range</i> = Not applicable<br><i>default</i> = Null                                                                                                                                                                                                                                                   |

|                                                                                       |                                                                                                                                               |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CCSRC</b><br>Target Address in FSC for the<br>Contact Cutout Source<br>  . . . . . | <b>Applies only if ALMOPT = Offnorml or CMDDIS and<br/>PNTFORM = Full</b><br><i>range</i> = 0 to 65535<br><i>default</i> = 0 (Not Configured) |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

## FSC-SM Digital Input Data Point

|                         |           |       |
|-------------------------|-----------|-------|
| FSC-SM DigIn Data Point | Engineer: | Date: |
|-------------------------|-----------|-------|

### Point Assignment Display

|                |           |                               |
|----------------|-----------|-------------------------------|
| <b>NAME</b>    |           | <i>range =</i> Not applicable |
| Tag Name       | . . . . . | <i>default =</i> Blank        |
| <b>NODETYP</b> | PM        | <i>range =</i> Not applicable |
| Node Type      | APM       | <i>default =</i> PM           |
|                | HPM       |                               |
|                | LM        |                               |
|                | <b>SM</b> |                               |

Applicable if NODETYP = SM

|                                   |             |                                                                                     |
|-----------------------------------|-------------|-------------------------------------------------------------------------------------|
| <b>SMPLTFM</b>                    | <b>FSC</b>  |                                                                                     |
| Type of Safety Control Platform   | TRICON      | Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller |
| <b>PNTFORM</b>                    | <b>Full</b> | Point is fully displayed and alarmed                                                |
| Point Form                        | Componnt    | Point is partially displayed but not alarmed                                        |
| <b>PTDESC</b>                     |             | <i>range =</i> Not applicable                                                       |
| Point Descriptor                  |             | <i>default =</i> Blank                                                              |
|                                   | . . . . .   |                                                                                     |
| <b>EUDESC</b>                     | . . . . .   | <i>range =</i> Not applicable                                                       |
| Engineering Units Descriptor      |             | <i>default =</i> Blank                                                              |
| <b>KEYWORD</b>                    | . . . . .   | <i>range =</i> Not applicable                                                       |
| Keyword Descriptor                |             | <i>default =</i> Blank                                                              |
| <b>ASSOCDSP</b>                   | . . . . .   | <i>range =</i> Alphanumeric                                                         |
| Associated Display                |             | <i>default =</i> Blank                                                              |
| <b>UNIT</b>                       | . .         | <i>range =</i> Alphanumeric                                                         |
| Process Unit Point is Assigned to |             | <i>default =</i> Not applicable                                                     |
| <b>NTWKNUM</b>                    | . .         | <i>range =</i> 1 to 20                                                              |
| NIM's UCN that This Point Is In   |             | <i>default =</i> Not applicable                                                     |
| <b>NODENUM</b>                    | . .         | <i>range =</i> 1 to 64                                                              |
| SMM's Address on the UCN          |             | <i>default =</i> Not applicable                                                     |
| <b>SLOTNUM</b>                    | . . . .     | <i>range =</i> 1 to 2000                                                            |
| Point Slot Number                 |             | <i>default =</i> Not applicable                                                     |

N/A if PNTFORM = Componnt

|                         |           |                               |
|-------------------------|-----------|-------------------------------|
| <b>PRIMMOD</b>          |           | <i>range =</i> Not applicable |
| Primary Module Point ID | . . . . . | <i>default =</i> Null         |

# FSC-SM Digital Input Data Point

*Point Assignment Display (continued)*

|                                                              |                                                                                 |                                     |
|--------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------|
| <b>DITYPE</b><br>Digital Input Type                          | <b>Status</b>                                                                   | Status input (open/closed contacts) |
| <b>PLCADDR</b><br>Source Address in FSC for the PVRAW input. | . . . . .  <br><i>range</i> = 0 to 65535<br><i>default</i> = 0 (Not Configured) |                                     |

## Operating Configuration Display

N/A if PNTFORM = Component

|                                                 |                            |                                                         |                                                                                                                                                                       |                  |
|-------------------------------------------------|----------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>STATETXT (1)</b><br>State 1 Description Text | . . . . .                  | <i>range</i> = Not applicable<br><i>default</i> = "On"  |                                                                                                                                                                       |                  |
| <b>STATETXT (0)</b><br>State 0 Description Text | . . . . .                  | <i>range</i> = Not applicable<br><i>default</i> = "Off" |                                                                                                                                                                       |                  |
| <b>BOXCLR (1)</b><br>State 1 Box Color          | Red<br>Cyan                | <b>Green</b> (Upper box default)<br>Yellow              | White<br>Blue                                                                                                                                                         | Black<br>Magenta |
| <b>BOXCLR (0)</b><br>State 0 Box Color          | Red<br>Cyan                | Green<br><b>Yellow</b> (Lower box default)              | White<br>Blue                                                                                                                                                         | Black<br>Magenta |
| <b>INPTDIR</b><br>Digital Input Direction       | <b>Direct</b><br>Reverse   |                                                         | Open => pvraw = off; Closed => pvraw = on<br>Closed => pvraw = off; Open => pvraw = on                                                                                |                  |
| <b>PVSRCOPT</b><br>PV Source Option             | <b>OnlyAuto</b><br><br>All |                                                         | PV source selection is not available and field wiring or memory fetch supplies PV.<br>Selection from PVSOURCE is available and Operator/CL-program may supply PV.     |                  |
| <b>PVSOURCE</b><br>PV Source                    | <b>Auto</b><br>Man<br>Sub  |                                                         | <b>Applies when PVSRCOPT = All.</b><br>Field wiring or memory fetch supplies PV.<br>PV is supplied by Operator or program.<br>A value is substituted by a CL program. |                  |

# FSC-SM Digital Input Data Point

## Alarming Display

N/A if PNTFORM = Component

| <b>\$AUXUNIT</b><br>Auxiliary Unit                                                                                                    | <input type="text"/>                                                          | range = Any configured unit in NCF<br>default = - -                                                                                                                                                                                                                                                                                      |                  |               |                        |     |     |     |    |     |     |    |    |    |
|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------|------------------------|-----|-----|-----|----|-----|-----|----|----|----|
| <b>\$REALARM</b><br>Re-alarming Interval (R660 and later)                                                                             | <input type="text"/>                                                          | <b>Applies when ALMOPT = Offnorml.</b><br>range = 0 to 9999<br>default = 0 minutes                                                                                                                                                                                                                                                       |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>ALMOPT</b><br>Alarming Option                                                                                                      | <input type="text"/>                                                          | 0 = None<br>1 = Offnorml<br>3 = Chngofst                                                                                                                                                                                                                                                                                                 |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>PVNORMAL</b><br>PV Value's Normal State Text                                                                                       | <input type="text"/>                                                          | <b>Applies when ALMOPT = Offnorml.</b><br>range = STATETXT(0) – STATETXT(1)<br>default = Text equal to "Off"                                                                                                                                                                                                                             |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>DLYTIME</b><br>Alarm Delay in Seconds                                                                                              | <input type="text"/>                                                          | <b>Applies when ALMOPT = Offnorml.</b><br>range = 0 to 60<br>default = 5 (seconds)                                                                                                                                                                                                                                                       |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>OFFNRMPR</b><br>Alarm Priority                                                                                                     | <b>NoAction</b><br>Journal<br>Low<br>High<br>Emergency<br>Printer<br>Jnlprint | <b>Applies when ALMOPT = Offnorml.</b><br>Alarm is not reported to the system. (R670 and later)<br>Logged but not reported to Op Stations.<br>Reported to Unit Alarm Summary display.<br>Reported to Area and Unit Alarm Summary.<br>Reported to all Alarm Summary displays.<br>Reported to US RTJ only<br>Reported to US RTJ and logged |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>ALENBST</b><br>Point Alarm Enable Status                                                                                           | <b>Enable</b><br>Disable<br>Inhibit                                           | <b>Applies when ALMOPT = Offnorml.</b><br><table> <tr> <th><u>Displayed</u></th> <th><u>Logged</u></th> <th><u>Reported to EIP</u></th> </tr> <tr> <td>Yes</td> <td>Yes</td> <td>Yes</td> </tr> <tr> <td>No</td> <td>Yes</td> <td>Yes</td> </tr> <tr> <td>No</td> <td>No</td> <td>No</td> </tr> </table>                                 | <u>Displayed</u> | <u>Logged</u> | <u>Reported to EIP</u> | Yes | Yes | Yes | No | Yes | Yes | No | No | No |
| <u>Displayed</u>                                                                                                                      | <u>Logged</u>                                                                 | <u>Reported to EIP</u>                                                                                                                                                                                                                                                                                                                   |                  |               |                        |     |     |     |    |     |     |    |    |    |
| Yes                                                                                                                                   | Yes                                                                           | Yes                                                                                                                                                                                                                                                                                                                                      |                  |               |                        |     |     |     |    |     |     |    |    |    |
| No                                                                                                                                    | Yes                                                                           | Yes                                                                                                                                                                                                                                                                                                                                      |                  |               |                        |     |     |     |    |     |     |    |    |    |
| No                                                                                                                                    | No                                                                            | No                                                                                                                                                                                                                                                                                                                                       |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>EVTOPT</b><br>Event Reporting Option                                                                                               | <input type="text"/>                                                          | <b>Applies only if ALMOPT = Offnorml or Chngofst.</b><br>0 = None<br>1 = Eip - Event Initiated Processing<br>2 = EipSoe - Event Initiated Proc. and Sequence of Events<br>3 = Soe - Sequence of Events                                                                                                                                   |                  |               |                        |     |     |     |    |     |     |    |    |    |
| <b>PVCHGDLY</b><br>Preset in seconds for the PV Change Delay function which throttles timestamped event collection. Refer to PVCHTMR. | <input type="text"/>                                                          | <b>Applies when EVTOPT = Soe or EipSoe.</b><br>range = 0 to 60<br>default = 0                                                                                                                                                                                                                                                            |                  |               |                        |     |     |     |    |     |     |    |    |    |

# FSC-SM Digital Input Data Point

Alarming Display (continued)

N/A if PNTFORM = Componnt

**EIPPCODE**

Event Init. Proc. Destination

| . . . . . |

**Applies when EVTOPT = Eip or EipSoe.**

*range* = Not applicable

*default* = Null

N/A if ALMOPT = None.

**CCSRC**

Target Address in FSC for the  
Contact Cutout Source

| . . . . . |

**Applies only if ALMOPT = Offnorml or Chngofst.**

*range* = 0 to 65535

*default* = 0 (Not Configured/UCN Access allowed for  
CONTCUT)

## FSC-SM Digital Output Data Point

|                          |           |       |
|--------------------------|-----------|-------|
| FSC-SM DigOut Data Point | Engineer: | Date: |
|--------------------------|-----------|-------|

### Point Assignment Display

|                |           |                               |
|----------------|-----------|-------------------------------|
| <b>NAME</b>    |           | <i>range =</i> Not applicable |
| Tag Name       | . . . . . | <i>default =</i> Blank        |
| <b>NODETYP</b> | PM        | <i>range =</i> Not applicable |
| Node Type      | APM       | <i>default =</i> PM           |
|                | HPM       |                               |
|                | LM        |                               |
|                | <b>SM</b> |                               |

Applicable if NODETYP = SM

|                                    |             |                                              |
|------------------------------------|-------------|----------------------------------------------|
| <b>SMPLTFM</b>                     | <b>FSC</b>  | Safety Manager is using FSC Controller       |
| Type of Safety Control Platform    | TRICON      | Safety Manager is using TRICON Controller    |
| <b>PNTFORM</b>                     | <b>Full</b> | Point is fully displayed and alarmed         |
| Point Form                         | Componnt    | Point is partially displayed but not alarmed |
| <b>PTDESC</b>                      |             | <i>range =</i> Not applicable                |
| Point Descriptor                   |             | <i>default =</i> Blank                       |
|                                    | . . . . .   |                                              |
| <b>EUDESC</b>                      | . . . . .   | <i>range =</i> Not applicable                |
| Engineering Units Descriptor       |             | <i>default =</i> Blank                       |
| <b>KEYWORD</b>                     | . . . . .   | <i>range =</i> Not applicable                |
| Point Keyword Descriptor           |             | <i>default =</i> Blank                       |
| <b>ASSOCDSP</b>                    | . . . . .   | <i>range =</i> Alphanumeric                  |
| Associated Display                 |             | <i>default =</i> Blank                       |
| <b>UNIT</b>                        | . .         | <i>range =</i> Alphanumeric                  |
| Process Unit Point is Assigned to  |             | <i>default =</i> Not applicable              |
| <b>NTWKNUM</b>                     | . .         | <i>range =</i> 1 to 20                       |
| NIM's UCN that Contains This Point |             | <i>default =</i> Not applicable              |
| <b>NODENUM</b>                     | . .         | <i>range =</i> 1 to 64                       |
| SMM's Address on the UCN           |             | <i>default =</i> Not applicable              |
| <b>SLOTNUM</b>                     | . . . .     | <i>range =</i> 1 to 2000                     |
| Point Slot Number                  |             | <i>default =</i> Not applicable              |

N/A if PNTFORM = Componnt

|                         |           |                               |
|-------------------------|-----------|-------------------------------|
| <b>PRIMMOD</b>          |           | <i>range =</i> Not applicable |
| Primary Module Point ID |           | <i>default =</i> Null         |
|                         | . . . . . |                               |

## FSC-SM Digital Output Data Point

### Point Assignment Display (continued)

|                                           |               |                                                                  |
|-------------------------------------------|---------------|------------------------------------------------------------------|
| <b>PLCADDR</b><br>FSC Destination Address | . . . . .     | <i>range</i> = 0 to 65535<br><i>default</i> = 0 (Not Configured) |
| <b>DOTYPE</b><br>Digital Output Type      | <b>Status</b> | Status input output<br>Pulse Width Modulated type output         |

### Operating Configuration Display

N/A if PNTFORM = Component

|                                                 |                                    |                                                                                                                     |               |                  |
|-------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|------------------|
| <b>STATETXT (1)</b><br>State 1 Description Text | . . . . .                          | <i>range</i> = Not applicable<br><i>default</i> = "On"                                                              |               |                  |
| <b>STATETXT (0)</b><br>State 0 Description Text | . . . . .                          | <i>range</i> = Not applicable<br><i>default</i> = "Off"                                                             |               |                  |
| <b>BOXCLR (1)</b><br>State 1 Box Color          | Red<br>Cyan                        | <b>Green</b> (Upper box default)<br>Yellow                                                                          | White<br>Blue | Black<br>Magenta |
| <b>BOXCLR (0)</b><br>State 0 Box Color          | Red<br>Cyan                        | Green<br><b>Yellow</b> (Lower box default)                                                                          | White<br>Blue | Black<br>Magenta |
| <b>NMODATTR</b><br>Normal Mode Attribute        | <b>None</b><br>Operator<br>Program | Parameter NMODE sets "normal" mode<br>Operator sets "normal" operating mode<br>Program sets "normal" operating mode |               |                  |
| <b>MODEPERM</b><br>Mode Permissive              | <b>Permit</b><br>NotPerm           | Operator can change this point's mode attribute<br>Operator cannot change this point's mode attribute               |               |                  |

## FSC-SM Logic Point

|                         |           |       |
|-------------------------|-----------|-------|
| FSC-SM Logic Data Point | Engineer: | Date: |
|-------------------------|-----------|-------|

### Point Assignment Display

|                 |           |                               |
|-----------------|-----------|-------------------------------|
| <b>NAME</b>     |           | <i>range =</i> Not applicable |
| Tag Name        | . . . . . | <i>default =</i> Blank        |
| <b>NODETYPE</b> | PM        | <i>range =</i> Not applicable |
| Node Type       | APM       | <i>default =</i> PM           |
|                 | HPM       |                               |
|                 | LM        |                               |
|                 | <b>SM</b> |                               |

|                                   |             |                                              |
|-----------------------------------|-------------|----------------------------------------------|
| Applicable if NODETYP = SM        |             |                                              |
| <b>SMPLTFM</b>                    | <b>FSC</b>  | Safety Manager is using FSC Controller       |
| Type of Safety Control Platform   | TRICON      | Safety Manager is using TRICON Controller    |
| <b>PNTFORM</b>                    | <b>Full</b> | Point is fully displayed and alarmed         |
| Point Form                        | Componnt    | Point is partially displayed but not alarmed |
| <b>PTDESC</b>                     |             | <i>range =</i> Not applicable                |
| Point Descriptor                  |             | <i>default =</i> Blank                       |
|                                   | . . . . .   |                                              |
| <b>ASSOCDSP</b>                   | . . . . .   | <i>range =</i> Alphanumeric                  |
| Associated Display                |             | <i>default =</i> Blank                       |
| <b>UNIT</b>                       | . .         | <i>range =</i> Alphanumeric                  |
| Process Unit Point is Assigned to |             | <i>default =</i> Not applicable              |
| <b>NTWKNUM</b>                    | . .         | <i>range =</i> 1 to 20                       |
| NIM's UCN that This Point Is In   |             | <i>default =</i> Not applicable              |
| <b>NODENUM</b>                    | . .         | <i>range =</i> 1 to 64                       |
| SMM's Address on the UCN          |             | <i>default =</i> Not applicable              |
| <b>SLOTNUM</b>                    | . .         | <i>range =</i> 1 to 30                       |
| Point Slot Number                 |             | <i>default =</i> Not applicable              |

Actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.

|                           |                                           |
|---------------------------|-------------------------------------------|
| N/A if PNTFORM = Componnt |                                           |
| <b>PRIMMOD</b>            | <i>range =</i> Not applicable             |
| Primary Module Point ID   | <i>default =</i> Null                     |
|                           | . . . . .                                 |
| <b>LOGMIX</b>             | <b>12_0_12</b>                            |
| Logic Mix                 | 12 inputs    0 logic blocks    12 outputs |

# FSC-SM Logic Point

*NIM Input Connections Display*

## LISRC

Source Addresses in FSC for the Logic Input

### Input

### Source Point.Parameter

|            |                       |
|------------|-----------------------|
| LISRC (1)  | . . . . .   . . . . . |
| LISRC (2)  | . . . . .   . . . . . |
| LISRC (3)  | . . . . .   . . . . . |
| LISRC (4)  | . . . . .   . . . . . |
| LISRC (5)  | . . . . .   . . . . . |
| LISRC (6)  | . . . . .   . . . . . |
| LISRC (7)  | . . . . .   . . . . . |
| LISRC (8)  | . . . . .   . . . . . |
| LISRC (8)  | . . . . .   . . . . . |
| LISRC (10) | . . . . .   . . . . . |
| LISRC (11) | . . . . .   . . . . . |
| LISRC (12) | . . . . .   . . . . . |

## LIBADOPT

Logic Input Failure Handling Option

Off

On

**Hold**

"Off" state is substituted for bad input.

"On" state is substituted for bad input.

Last good value is substituted for bad input.

## LODSTN

Destination Addresses in FSC for the Logic output

### Output

### Destination Point.Parameter

|             |                       |
|-------------|-----------------------|
| LODSTN (1)  | . . . . .   . . . . . |
| LODSTN (2)  | . . . . .   . . . . . |
| LODSTN (3)  | . . . . .   . . . . . |
| LODSTN (4)  | . . . . .   . . . . . |
| LODSTN (5)  | . . . . .   . . . . . |
| LODSTN (6)  | . . . . .   . . . . . |
| LODSTN (7)  | . . . . .   . . . . . |
| LODSTN (8)  | . . . . .   . . . . . |
| LODSTN (9)  | . . . . .   . . . . . |
| LODSTN (10) | . . . . .   . . . . . |
| LODSTN (11) | . . . . .   . . . . . |
| LODSTN (12) | . . . . .   . . . . . |

# FSC-SM Logic Point

*NIM Input Connections Display (continued)*

## LOENBL

Source Address in FSC for enable

### Enable

### PLC Address

|             |         |                                                 |
|-------------|---------|-------------------------------------------------|
| LOENBL (1)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (2)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (3)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (4)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (5)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (6)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (7)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (8)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (9)  | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (10) | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (11) | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |
| LOENBL (12) | . . . . | <i>range = 0 to 65535</i><br><i>default = 0</i> |

## FSC-SM Logic Point

*NIM Logic Descriptors Display*

N/A if PNTFORM = Component

**NODESC**

Number of Generic Descriptors

*range* = 0 to 12

*default* = 0

**GENDESC**

Generic Descriptors (defaults = blanks)

GENDESC(1)

GENDESC(2)

A number line from 0 to 10. Tick marks are present at every integer. The segment between 4 and 5 is highlighted with a bracket and labeled  $\frac{4}{10}$ .

GENDESC(3)

| . |

---

| . |

---

GENDESC(4)

GENDESC(5)

\_\_\_\_\_

GENDESC(6)

GENDESC(7)

A horizontal number line is shown, starting at 0 and ending at 10. Tick marks are placed at every integer from 0 to 10. The segment of the number line between the numbers 4 and 5 is highlighted with a thick black line.

GENDESC(8)

A horizontal number line with vertical tick marks at each integer from 0 to 10. The segment between the tick marks for 4 and 5 is shaded gray.

GENDESC(9)

\_\_\_\_\_

GENDESC(10)

A horizontal number line is shown with vertical tick marks at each integer from 0 to 10. The segment of the line between the tick marks for 4 and 5 is shaded gray.

GENDESC(11)

A number line from 0 to 10. Tick marks are present at every integer. The segment between 4 and 5 is highlighted with a bracket and labeled  $\frac{4}{10}$ .

GENDESC(12)

A horizontal number line with vertical tick marks at each integer from 0 to 10. The numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are written below the corresponding tick marks. The tick mark for the number 5 is circled.

## FSC-SM Flag Point

|                        |           |       |
|------------------------|-----------|-------|
| FSC-SM Flag Data Point | Engineer: | Date: |
|------------------------|-----------|-------|

### Point Assignment Display

|                             |                                     |                                                         |
|-----------------------------|-------------------------------------|---------------------------------------------------------|
| <b>NAME</b><br>Tag Name     | <input type="text"/>                | <i>range</i> = Not applicable<br><i>default</i> = Blank |
| <b>NODETYP</b><br>Node Type | PM<br>APM<br>HPM<br>LM<br><b>SM</b> | <i>range</i> = Not applicable<br><i>default</i> = PM    |

Applicable if NODETYP = SM

|                                                   |                         |                                                                                      |
|---------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------|
| <b>SMPLTFM</b><br>Type of Safety Control Platform | <b>FSC</b><br>TRICON    | Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller  |
| <b>PNTFORM</b><br>Point Form                      | <b>Full</b><br>Componnt | Point is fully displayed and alarmed<br>Point is partially displayed but not alarmed |
| <b>PTDESC</b><br>Point Descriptor                 | <input type="text"/>    | <i>range</i> = Not applicable<br><i>default</i> = Blank                              |
| <b>EUDESC</b><br>Engineering Units Descriptor     | <input type="text"/>    | <i>range</i> = Not applicable<br><i>default</i> = Blank                              |
| <b>KEYWORD</b><br>Point Keyword                   | <input type="text"/>    | <i>range</i> = Not applicable<br><i>default</i> = Blank                              |
| <b>ASSOCDSP</b><br>Associated Display             | <input type="text"/>    | <i>range</i> = Alphanumeric<br><i>default</i> = Blank                                |
| <b>UNIT</b><br>Process Unit Point is Assigned to  | <input type="text"/>    | <i>range</i> = Alphanumeric<br><i>default</i> = Not applicable                       |
| <b>NTWKNUM</b><br>NIM's UCN that This Point Is In | <input type="text"/>    | <i>range</i> = 1 to 20<br><i>default</i> = Not applicable                            |
| <b>NODENUM</b><br>SMM's Address on the UCN        | <input type="text"/>    | <i>range</i> = 1 to 64<br><i>default</i> = Not applicable                            |
| <b>SLOTNUM</b><br>Point Slot Number               | <input type="text"/>    | <i>range</i> = 1 to 2000<br><i>default</i> = Not applicable                          |

The actual Range check will be calculated on the mix of configured Points desired.

N/A if PNTFORM = Componnt

|                                           |                      |                                                        |
|-------------------------------------------|----------------------|--------------------------------------------------------|
| <b>PRIMMOD</b><br>Primary Module Point ID | <input type="text"/> | <i>range</i> = Not applicable<br><i>default</i> = Null |
|-------------------------------------------|----------------------|--------------------------------------------------------|

## FSC-SM Flag Point

### Operating Configuration Display

N/A if PNTFORM = Componnt

|                                           |             |                                                       |
|-------------------------------------------|-------------|-------------------------------------------------------|
| <b>STATETXT (1)</b><br>State 1 Descriptor | . . . . .   | <i>range</i> = Not applicable<br><i>default</i> = ON  |
| <b>STATETXT (0)</b><br>State 0 Descriptor | . . . . .   | <i>range</i> = Not applicable<br><i>default</i> = OFF |
| <b>BOXCLR (1)</b><br>State 1 Box Color    | Red<br>Cyan | Green<br><b>Yellow</b>                                |
|                                           |             | White<br>Blue                                         |
|                                           |             | Black<br>Magenta                                      |
| <b>BOXCLR (0)</b><br>State 0 Box Color    | Red<br>Cyan | <b>Green</b><br>Yellow                                |
|                                           |             | White<br>Blue                                         |
|                                           |             | Black<br>Magenta                                      |

### Alarming Display

N/A if PNTFORM = Componnt or SLOTNUM > 512

|                                                                        |                                                                               |                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>\$AUXUNIT</b>                                                       |                                                                               | <i>range</i> = Any configured unit in NCF<br><i>default</i> = - -                                                                                                                                                                                                                                      |
| <b>\$REALARM</b><br>Re-alarming Interval (R660 and later)              |                                                                               | <i>range</i> = 0 to 9999 minutes<br><i>default</i> = 0 minutes                                                                                                                                                                                                                                         |
| <b>OFFNRMPR</b><br>Off Normal Alarm Priority                           | <b>NoAction</b><br>Journal<br>Low<br>High<br>Emergency<br>Printer<br>Jnlprint | Alarm is not reported to the system. (R670 and later)<br>Logged but not reported to Operator Stations.<br>Reported to Unit Alarm Summary display.<br>Reported to Area and Unit Alarm Summary.<br>Reported to all Alarm Summary displays.<br>Reported to US RTJ only.<br>Reported to US RTJ and logged. |
| <b>ALENBST</b><br>Alarm Enable State                                   | <b>Enable</b><br>Disable<br>Inhibit                                           | <i>default</i> = Enable                                                                                                                                                                                                                                                                                |
| <b>EIPPCODE</b><br>EIP Point ID                                        | . . . . .                                                                     | <i>range</i> = Not applicable<br><i>default</i> = Blank                                                                                                                                                                                                                                                |
| <b>CCSRC</b><br>Target Address in FSC for the<br>Contact Cutout Source | . . . . .                                                                     | <i>range</i> = 0 to 65535<br><i>default</i> = 0                                                                                                                                                                                                                                                        |

## FSC-SM Timer Point

|                                |                  |              |
|--------------------------------|------------------|--------------|
| FSC-SM <b>Timer</b> Data Point | <b>Engineer:</b> | <b>Date:</b> |
|--------------------------------|------------------|--------------|

### Point Assignment Display

|                |           |                               |
|----------------|-----------|-------------------------------|
| <b>NAME</b>    |           | <i>range =</i> Not applicable |
| Tag Name       | . . . . . | <i>default =</i> Blank        |
| <b>NODETYP</b> |           | <i>range =</i> Not applicable |
| Node Type      | PM        | <i>default =</i> PM           |
|                | APM       |                               |
|                | HPM       |                               |
|                | LM        |                               |
|                | <b>SM</b> |                               |

Applicable if NODETYP = SM

|                                    |             |                                                                                      |
|------------------------------------|-------------|--------------------------------------------------------------------------------------|
| <b>SMPLTFM</b>                     | <b>FSC</b>  |                                                                                      |
| Type of Safety Control Platform    | TRICON      | Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller  |
| <b>PNTFORM</b>                     | <b>Full</b> |                                                                                      |
| Point Form                         | Componnt    | Point is fully displayed and alarmed<br>Point is partially displayed but not alarmed |
| <b>PTDESC</b>                      |             | <i>range =</i> Not applicable                                                        |
| Point Descriptor                   | . . . . .   | <i>default =</i> Blank                                                               |
| <b>EUDESC</b>                      | . . . . .   | <i>range =</i> Not applicable                                                        |
| Engineering Units Descriptor       |             | <i>default =</i> Blank                                                               |
| <b>KEYWORD</b>                     | . . . . .   | <i>range =</i> Not applicable                                                        |
| Keyword Descriptor                 |             | <i>default =</i> Blank                                                               |
| <b>ASSOCDSP</b>                    | . . . . .   | <i>range =</i> Alphanumeric                                                          |
| Associated Display                 |             | <i>default =</i> Blank                                                               |
| <b>UNIT</b>                        | .           | <i>range =</i> Alphanumeric                                                          |
| Process Unit Point is Assigned to  |             | <i>default =</i> Not applicable                                                      |
| <b>NTWKNUM</b>                     | .           | <i>range =</i> 1 to 20                                                               |
| NIM's UCN that Contains This Point |             | <i>default =</i> Not applicable                                                      |
| <b>NODENUM</b>                     | .           | <i>range =</i> 1 to 64                                                               |
| SMM's Address on the UCN           |             | <i>default =</i> Not applicable                                                      |
| <b>SLOTNUM</b>                     | . . .       | <i>range =</i> 1 to 1500                                                             |
| Point Slot Number                  |             | <i>default =</i> Not applicable                                                      |

The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.

N/A if PNTFORM = Componnt

|                         |                               |
|-------------------------|-------------------------------|
| <b>PRIMMOD</b>          | <i>range =</i> Not applicable |
| Primary Module Point ID | <i>default =</i> Null         |
| . . . . .               |                               |

## FSC-SM Timer Point

### *Point Assignment Display (Continued)*

|                                                                                                                    |         |                                     |
|--------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------|
| <b>PLCADDR</b>                                                                                                     | . . . . | <i>range = 0 to 65535</i>           |
| Target Address in FSC for the Set/Reset Contact                                                                    |         | <i>default = 0 (Not Configured)</i> |
| <b>The PLCADDR for the reset contact in FSC must be equal to the PLCADDR for the set contact incremented by 1.</b> |         |                                     |

|                                                 |         |                                     |
|-------------------------------------------------|---------|-------------------------------------|
| <b>PVSLTSRC</b>                                 | . . . . | <i>range = 0 to 65535</i>           |
| Target Address in FSC for the Accumulator Value |         | <i>default = 0 (Not Configured)</i> |

|                                            |         |                                     |
|--------------------------------------------|---------|-------------------------------------|
| <b>SPSLTSRC</b>                            | . . . . | <i>range = 0 to 65535</i>           |
| Target Address in FSC for the Preset Value |         | <i>default = 0 (Not Configured)</i> |

### *NIM Operating Configuration Display*

|                   |           |    |    |
|-------------------|-----------|----|----|
| <b>PVFORMAT</b>   | <b>D0</b> | D1 | D2 |
| PV Decimal Format |           |    |    |

## FSC-SM Numeric Point

|                           |           |       |
|---------------------------|-----------|-------|
| FSC-SM Numeric Data Point | Engineer: | Date: |
|---------------------------|-----------|-------|

### Point Assignment Display

|                             |                                     |                                                         |
|-----------------------------|-------------------------------------|---------------------------------------------------------|
| <b>NAME</b><br>Tag Name     | <input type="text"/>                | <i>range</i> = Not applicable<br><i>default</i> = Blank |
| <b>NODETYP</b><br>Node Type | PM<br>APM<br>HPM<br>LM<br><b>SM</b> | <i>range</i> = Not applicable<br><i>default</i> = PM    |

|                                                      |                         |                                                                                      |
|------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------|
| <i>Applicable if NODETYP = SM</i>                    |                         |                                                                                      |
| <b>SMPLTFM</b><br>Type of Safety Control Platform    | <b>FSC</b><br>TRICON    | Safety Manager is using FSC Controller<br>Safety Manager is using TRICON Controller  |
| <b>PNTFORM</b><br>Point Form                         | <b>Full</b><br>Componnt | Point is fully displayed and alarmed<br>Point is partially displayed but not alarmed |
| <b>PTDESC</b><br>Point Descriptor                    |                         | <i>range</i> = Not applicable<br><i>default</i> = Blank                              |
| <input type="text"/>                                 |                         |                                                                                      |
| <b>EUDESC</b><br>Engineering Units Descriptor        | <input type="text"/>    | <i>range</i> = Not applicable<br><i>default</i> = Blank                              |
| <b>KEYWORD</b><br>Keyword Descriptor                 | <input type="text"/>    | <i>range</i> = Not applicable<br><i>default</i> = Blank                              |
| <b>ASSOCDSP</b><br>Associated Display                | <input type="text"/>    | <i>range</i> = Alphanumeric<br><i>default</i> = Blank                                |
| <b>UNIT</b><br>Process Unit Point is Assigned to     | <input type="text"/>    | <i>range</i> = Alphanumeric<br><i>default</i> = Not applicable                       |
| <b>NTWKNUM</b><br>NIM's UCN that Contains This Point | <input type="text"/>    | <i>range</i> = 1 to 20<br><i>default</i> = Not applicable                            |
| <b>NODENUM</b><br>SMM's Address on the UCN           | <input type="text"/>    | <i>range</i> = 1 to 64<br><i>default</i> = Not applicable                            |
| <b>SLOTNUM</b><br>Point Slot Number                  | <input type="text"/>    | <i>range</i> = 1 to 1000<br><i>default</i> = Not applicable                          |

The actual Range check will be calculated on the mix of configured Points desired.

## FSC-SM Numeric Point

*Point Assignment Display (Continued)*

|                         |           |                           |                   |
|-------------------------|-----------|---------------------------|-------------------|
|                         |           | N/A if PNTFORM = Componnt |                   |
| <b>PRIMMOD</b>          |           | <i>range =</i>            | Not applicable    |
| Primary Module Point ID |           | <i>default =</i>          | Null              |
|                         | . . . . . |                           |                   |
| <b>PVFORMAT</b>         | .         | <i>range:</i>             | D0 (XXXX.)        |
|                         |           |                           | <b>D1</b> (XXX.X) |
|                         |           |                           | D2 (XX.XX)        |
|                         |           |                           | D3 (X.XXX)        |
|                         |           | <i>default:</i>           | D1                |



